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MASTER OF SCIENCE DISSERTATION

Dissertation Title:	SENTIMENT ANALYSIS OF STOCK MARKET USING FINANCIAL NEWS ARTICLES
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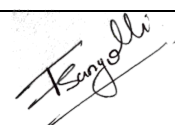
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Glossary

- **AAPL:** Apple Inc. (stock ticker symbol for Apple)
- **CSV:** Comma-Separated Values (a file format used to store tabular data)
- **LSTM:** Long Short-Term Memory (a type of recurrent neural network used in deep learning)
- **ML:** Machine Learning (a subset of artificial intelligence focused on data-driven algorithms)
- **NLP:** Natural Language Processing (a field of AI that focuses on the interaction between computers and human language)
- **SVM:** Support Vector Machine (a supervised machine learning algorithm used for classification tasks)
- **VADER:** Valence Aware Dictionary and sEntiment Reasoner (a lexicon-based sentiment analysis model)
- **Text Blob:** A Python library used for processing textual data and sentiment analysis

Abstract

This project explores the relationship between financial news sentiment and stock price movements using web scraping and sentiment analysis techniques. Focusing on Apple Inc. (AAPL) as a case study, financial news headlines were scraped from Finviz and analysed using sentiment analysis model VADER. The project aimed to determine how sentiment influences stock performance, particularly during significant events like the release of the iPhone 16.

A Python-based web scraping tool was developed to automate the collection of news headlines, which were then processed to classify sentiment as positive, neutral, or negative. VADER and Text Blob models were compared, with Text Blob achieving a slightly higher accuracy (58.53%) than VADER (55.03%). Despite this, both models demonstrated a stronger ability to classify neutral sentiments compared to positive or negative sentiments.

The analysis revealed that stock price trends were generally aligned with sentiment shifts. Positive sentiment typically corresponded with price increases, while negative sentiment often preceded price drops. Notably, the release of the iPhone 16 on September 20, 2024, demonstrated a clear positive impact on Apple's stock price, despite mixed sentiment.

This project demonstrates the practical application of sentiment analysis in financial markets, offering insights into how public opinion can influence stock performance. It also highlights opportunities for future enhancements, such as integrating advanced machine learning models and extending the analysis to real-time data.

Keywords: sentiment analysis, financial markets, web scraping, VADER, Text Blob, stock price prediction, Apple Inc., Python

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1. Introduction

1.1 Background

In financial markets, the ability to quickly assess and respond to real-time news can provide a substantial competitive advantage for investors. Financial news articles, press releases, and media coverage significantly influence market sentiment, which directly impacts stock prices. Traditionally, investors and analysts have relied on manual reading and interpretation of these news sources, a process that is not only labour-intensive but also subject to human biases, potentially leading to suboptimal decisions.

Recent advancements in Natural Language Processing (NLP) have provided a solution in the form of automated sentiment analysis. Sentiment analysis allows computers to systematically evaluate the emotional tone of textual data, offering a scalable and reliable way to gauge public sentiment towards financial assets. For instance, the tone of news articles about a particular stock can be used to predict whether the stock is likely to rise or fall, based on how positive or negative the sentiment is.

However, the financial domain presents unique challenges for sentiment analysis. Financial language is often nuanced, and the emotional weight of words can vary depending on the context. While there are many general-purpose sentiment analysis tools available, only a few have been adapted for the specific needs of financial markets. This project seeks to address this gap by applying sentiment analysis techniques, specifically VADER (Valence Aware Dictionary and sEntiment Reasoner), to financial news scraped from the Finviz platform. By leveraging this model, the project aims to develop a robust system that automates the process of sentiment evaluation and assesses how sentiment influences stock price movements.

1.2 Aim

The primary aim of this project is to design and implement a system that scrapes financial news from the Finviz website at regular intervals, processes the data for sentiment analysis using VADER, and evaluates the relationship between news sentiment and stock market performance. Additionally, the project aims to validate the accuracy of VADER by comparing its performance with other sentiment analysis

tools, such as Text Blob, using an external dataset. The end goal is to provide visualizations that offer investors actionable insights into how news sentiment may impact stock prices.

1.3 Objectives

To achieve the above aim, the following objectives are defined:

- **Web Scraping:** Develop a Python-based tool that automatically scrapes financial news from Finviz daily. The data, consisting of the stock ticker, date, time, and headline, will be stored in CSV format for further analysis.
- **Data Pre-processing:** Implement a series of text pre-processing steps, such as removing unnecessary characters, dropping duplicates, sorting and formatting date for sentiment analysis.
- **Sentiment Analysis:** Apply VADER to classify news headlines into positive, negative, or neutral sentiments based on the compound score generated by the sentiment analyser.
- **Stock Performance Analysis:** Retrieve stock performance data (open, close, high, low prices) from Yahoo Finance and correlate the sentiment of the news with changes in stock prices on the same day.
- **Accuracy Validation:** Compare the accuracy of the VADER model with another sentiment analysis tool, Text Blob, by testing the models against a manually labelled dataset from Kaggle.
- **Visualization:** Provide graphical visualizations, such as line charts and bar graphs, to illustrate sentiment trends over time and how these trends correlate with stock performance.

1.4 Relevance and Significance

This project is highly relevant in the modern financial landscape, where the speed at which information is processed can mean the difference between profit and loss. With the increasing volume of financial news and the complexity of financial language, manually tracking sentiment is no longer feasible. Automated sentiment analysis provides a scalable solution, enabling investors to quickly understand the market sentiment and make informed decisions. By using a domain-specific sentiment analysis tool like VADER, which has been designed to handle nuanced language, this project addresses the challenges associated with analysing financial text and provides a practical application for investors, analysts, and researchers.

Through this system, investors will gain access to sentiment insights that can guide their trading decisions, allowing them to respond quickly to emerging news trends. This project not only contributes to the field of sentiment analysis but also offers a practical tool for financial market participants.

2. Literature Review

Sentiment analysis has emerged as a powerful tool across various domains, from financial markets to social media platforms. By analysing public opinion, this technique helps predict market movements, assess public policy reactions, and improve product recommendations. This literature review focuses on applying sentiment analysis in financial markets, social media and political sentiment, and improved lexicon-based approaches, followed by sector-specific applications. The review critically appraises the existing research, highlighting strengths, limitations, and areas for future development.

2.1 Sentiment Analysis in Financial Markets

One of the most common applications of sentiment analysis is predicting stock market trends, where public sentiment can significantly influence investor behaviour.

Bhardwaj et al. (2015) examine the Indian stock market, using data from Sensex and Nifty to predict market conditions through sentiment analysis. Their approach utilizes web scraping tools to collect live news data, which is then analysed to provide market predictions. While the study demonstrates the potential of real-time sentiment analysis in financial decision-making, its reliance on a single news source (Timesofindia.com) significantly limits its scope. The absence of multiple data sources and advanced machine learning models diminishes the robustness of the predictions, making the model potentially unreliable for high-stakes investment decisions. Expanding the news sources and integrating sophisticated machine learning models could provide a more accurate and resilient predictive system.

Li, Wu and Wang (2020) develop a two-layer LSTM neural network model that integrates stock prices with sentiment analysis from financial news articles. Their

model significantly outperforms traditional price-only models, highlighting the value of sentiment data in stock market predictions. However, their study's focus on a single financial market (Hong Kong) limits its broader applicability. A model validated on multiple markets would offer more generalized insights, making it useful for international investors. The study could also benefit from a deeper examination of potential overfitting issues in LSTM models when applied to small or noisy datasets. Future research should explore global markets and compare results across different economic environments.

Yu, Kang and Park (2019) take a different approach by analysing the impact of information availability and user opinions on Bitcoin's return volatility. Their study uses the GJR-GARCH model to evaluate how user interest, as measured by Google Trends, influences Bitcoin volatility. While this approach is novel, the study's focus on volatility asymmetry limits its ability to provide a holistic view of market efficiency. Moreover, the reliance on proxies like Google Trends and Wikipedia raises concerns about the representativeness of user sentiment, as these platforms may not capture the full spectrum of market influencers. A broader range of sentiment sources, including financial forums and blockchain-specific communities, would provide a more accurate depiction of market sentiment.

Shah, Isah and Zulkernine (2018) investigate the impact of news sentiments on stock prices in the pharmaceutical sector using a dictionary-based sentiment analysis model. Their sector-specific sentiment dictionary yields a notable 70.59% directional accuracy. However, the study's reliance on manually created dictionaries makes it difficult to scale across industries and markets. The time-consuming nature of manual dictionary construction also introduces a risk of outdated sentiment models. Automating the dictionary-building process through machine learning algorithms and expanding the model to other industries would enhance both its scalability and relevance.

2.2 Social Media and Political Sentiment Analysis

Social media platforms like Twitter and Reddit have become key sources for sentiment analysis, particularly in areas such as political discourse, where real-time public opinion is crucial.

Pak and Paroubek (2010) present one of the earlier studies to use Twitter as a corpus for sentiment analysis. They rely on emoticons to categorize tweets into positive, negative, and neutral sentiments, and apply a Naive Bayes classifier for classification. While the study successfully demonstrates the value of Twitter data for binary sentiment classification, its reliance on emoticons for sentiment inference overlooks the more nuanced sentiments expressed in text without emoticons. Furthermore, the study is limited to English tweets, raising concerns about its cross-linguistic applicability. Expanding the dataset to include tweets from multiple languages and applying more sophisticated sentiment extraction methods could enhance the study's relevance and accuracy.

Wang, Niu and Yu (2019) offer an innovative approach with their SentiDiff model, which integrates sentiment diffusion patterns with textual analysis. By analysing how sentiments spread through retweets, SentiDiff provides more accurate predictions than traditional text-only models. However, the focus on Twitter-specific dynamics limits the model's applicability to other social media platforms, such as Facebook or Reddit, which have different user behaviours and interaction mechanisms. Future studies should extend the sentiment diffusion model to other platforms and explore how sentiment spreads differently across these ecosystems.

Kaur (2022) applies machine learning algorithms to analyse political sentiment on Reddit, focusing on threads related to political news. The study compares several models, with logistic regression slightly outperforming Naive Bayes in accuracy. While the study successfully demonstrates the use of machine learning for political sentiment analysis, its narrow focus on Reddit's political threads may not reflect broader sentiment trends in other types of media. Additionally, the modest accuracy of the models (around 61%) suggests room for improvement by incorporating more advanced NLP techniques, such as transformer-based models.

2.3 Improved Lexicon-Based Approaches for Sentiment Analysis

Lexicon-based approaches continue to be widely used for sentiment analysis due to their simplicity and interpretability. However, recent advancements focus on improving these models to enhance their performance in more complex settings.

Bonta, Kumaresh and Janardhan (2019) compare three popular lexicon-based sentiment analysis tools: NLTK, Text Blob, and VADER. Their findings suggest that VADER performs better than the other tools in analysing social media content, primarily due to its ability to account for factors such as punctuation, capitalization, and slang. However, the study's narrow focus on movie reviews limits its conclusions, as different domains (such as political content or product reviews) may require different sentiment analysis approaches. Expanding the study to include a variety of datasets would allow for a more generalized assessment of these tools' effectiveness.

Veena, Vinayak and Nair (2021) propose an improvement to VADER by integrating dependency parsing, which allows for a more accurate analysis of sentences containing verbs. Their model achieves a slight improvement in accuracy over the original VADER algorithm, showing that verbs significantly contribute to sentiment interpretation. Despite the improvement, the study's scope is limited to analysing the Farm Bills Act, a specific political event in India. Broader applications of this improved VADER model, particularly in different languages and contexts, would help verify its generalizability.

Usha and Devi (2013) present the Combined Sentiment-Topic (CST) model, which combines sentiment analysis with topic detection using an unsupervised learning approach. The CST model demonstrates impressive results in analysing product reviews, though it struggles with classifying neutral sentiments, focusing mainly on positive and negative categories. Adding a mechanism to handle neutral sentiment, along with incremental learning capabilities, would significantly improve the model's robustness.

2.4 Sector-Specific Applications of Sentiment Analysis

Sentiment analysis has been successfully applied in various sectors beyond finance and social media, including health, education, and language-specific domains.

Mokhosi, Shivachi and Sethobane (2024) introduce a sentiment analysis dataset for Sesotho, an underrepresented African language. The dataset, which consists of

news headlines, is designed for both sentiment and aspect-based analysis, providing a valuable resource for African NLP research. However, the dataset is skewed towards negative sentiment, reflecting its focus on crime-related news, which limits its applicability to more balanced or positive datasets. Expanding the dataset to cover a broader range of topics would make it more versatile for general sentiment analysis in low-resource languages.

Sandra and Aritonang (2021) apply sentiment analysis to Twitter data related to Indonesia's PPKM lockdown during the COVID-19 pandemic. Using Artificial Neural Networks, the study finds that neutral sentiments dominate public opinion. A key limitation is the focus on a single platform (Twitter) and the predominance of neutral sentiment, which could obscure more specific emotional responses. Extending the analysis to other platforms or refining the sentiment categories could reveal more detailed insights into public sentiment during the lockdown.

Feliciani et al. (2024) focus on how crowd accidents are reported in the media, using sentiment analysis to examine the evolution of language used in these reports. Their findings highlight how sensationalist language, particularly fear-related terms, is commonly used in crowd accident reporting. A limitation of this study is the cultural narrowness of the dataset, which focuses primarily on Western and South Asian media sources. Incorporating a more globally representative dataset would provide more generalizable insights into how crowd accidents are reported worldwide.

3. Methodology

The methodology of this project is divided into three main parts: (1) the creation of a GitHub repository for automating data collection, (2) the processing and sentiment analysis of financial news using VADER, and (3) the accuracy evaluation of sentiment analysis models using external datasets. Each part plays a crucial role in ensuring the system operates efficiently and produces accurate sentiment analysis results.

3.1 Repository Setup and Data Collection

The first part of the project involved setting up an automated data collection system using GitHub Actions to scrape financial news articles from the Finviz website. The goal was to collect financial news headlines at regular intervals to perform sentiment analysis and track stock trends based on public sentiment.

- **Automation with GitHub Actions:** The project utilized GitHub Actions, a continuous integration and automation tool, to automate the daily execution of the Python script. The YAML file was configured to schedule the scraping job to run every day at 11:00 AM. This allowed GitHub Actions to automatically execute the script without any manual intervention, ensuring that the latest financial news was consistently retrieved and saved. The script employed Python's requests library to send HTTP requests to the Finviz website, fetching raw HTML data. This data was then parsed using BeautifulSoup, which extracted key information such as financial headlines, tickers, dates, and times. This setup provided a hands-off, scalable, and reliable solution for collecting financial news articles.
- **Data Storage:** After scraping the news articles, the data was stored in CSV format, with each file representing one day's news. The CSV files included four key columns: Ticker, Date, Time, and Title. The Ticker column identified the relevant stock, while the Title column captured the news headline. The Date and Time columns recorded when the news was published. This structure made the data easy to read and manipulate during the later stages of the project. Additionally, the repository was automatically updated daily with new CSV files, ensuring an organized and continuously up-to-date record of financial news.

- **Version Control:** Version control played a crucial role in this part of the project. With GitHub Actions, every time new data was scraped, it was automatically committed to the GitHub repository. This ensured that any changes or updates to the script, as well as the daily collected data, were properly tracked and documented. The use of GitHub's version control also facilitated the smooth integration of future modifications or improvements to the script.

3.2 Data Processing and Sentiment Analysis

The second stage of the project involved cleaning and processing the data collected from the Finviz website, followed by sentiment analysis using the VADER sentiment analysis tool. This step ensured the data was properly structured and ready for sentiment classification.

- **Data Pre-processing:** The first step in this stage involved downloading the entire repository as a ZIP file, which contained all the previously collected CSV files. A Python script was developed to read and merge these CSV files into a single Pandas Data Frame, making it easier to process the data as a whole rather than working with individual files. Several pre-processing steps were applied to clean and prepare the data for analysis:
 - **Removing Invalid Dates:** Some rows in the dataset contained the string 'today' in the Date column, which was not a valid format for analysis. These rows were removed to maintain the integrity of the dataset.
 - **Removing Duplicates:** Duplicate rows were identified and dropped to prevent redundant information from skewing the sentiment analysis results. This ensured that each news headline was only processed once.
 - **Reformatting Dates:** The Date column was converted into a valid date time format, which allowed for accurate sorting and filtering based on time. This step was crucial for analysing sentiment over time, as it ensured that the data was chronologically accurate.
 - **Sorting:** The data was sorted by Ticker and Date, ensuring that each stock's news was processed in the correct chronological order. This sorting facilitated a clear understanding of how sentiment evolved for each stock over time.

- **Sentiment Analysis:** After cleaning the data, sentiment analysis was performed using the VADER (Valence Aware Dictionary and sEntiment Reasoner) tool. VADER is a lexicon-based model specifically designed for short texts, making it ideal for analysing the sentiment of news headlines. VADER calculates a compound sentiment score for each headline, which ranges from -1 (most negative) to +1 (most positive). Based on this score, each news headline was classified as either positive, neutral, or negative. VADER was chosen because it is highly effective for real-time sentiment analysis and does not require extensive training data, making it well-suited for this project's focus on short, daily news headlines(sentiment analysis code).
- **Data Output:** After the sentiment analysis was completed, the resulting dataset, including the original headlines and their associated sentiment scores, was saved as a dataset file. This final dataset was then used for further exploration and visualization. Matplotlib and Seaborn libraries were employed to create visualizations, such as line graphs, that showed the distribution of sentiment over time. These visualizations helped illustrate the correlation between sentiment and stock performance.

3.3 Accuracy Evaluation and Model Comparison

The final part of the project focused on evaluating the accuracy of the sentiment analysis models, namely VADER and Text Blob, using a labelled dataset obtained from [Kaggle dataset](#). This step was crucial for comparing how well each model performed in classifying sentiment correctly.

- **Dataset Overview:** The Kaggle dataset included over 6,000 stock market-related news headlines, each labelled as positive, negative, or neutral. It covered a range of financial sources, making it ideal for testing the models across various news types. The dataset was balanced, with an even distribution of sentiment categories, ensuring a fair evaluation of both models.
- **Sentiment Classification:** Both VADER and Text Blob were employed to classify the sentiment of each news headline in the dataset. VADER relied on its compound score to assign sentiment categories, while Text Blob used its polarity score to determine whether a headline was positive, negative, or

neutral. This approach enabled a comparative evaluation of the two models, assessing how accurately they aligned with the true sentiment labels.

- **Accuracy Calculation:** The accuracy of each model was calculated by comparing the predicted sentiment to the true labels in the dataset. VADER achieved an accuracy of 55.03%, Text Blob, which achieved 58.53%. Although Text Blob performed slightly better results. But VADER demonstrated superior performance in accurately classifying positive, and negative sentiments.
- **Confusion Matrix:** To evaluate the performance of both models in greater detail, confusion matrices were generated. These matrices offered a comprehensive breakdown of true positives, false positives, true negatives, and false negatives for each sentiment class. Visualized using Seaborn, the results showed that VADER performed well in identifying both positive and negative sentiments, whereas Text Blob faced more challenges, particularly in accurately classifying neutral sentiments.

3.4 Technology Stack

The following tools and technologies were utilized in this project:

- **Python:** The primary programming language for developing the scraping script, data processing, and sentiment analysis.
- **BeautifulSoup** and **requests:** Used for web scraping and retrieving financial news articles from Finviz.
- **Pandas:** Essential for data manipulation, merging, and cleaning tasks in the pre-processing stage.
- **VADER** and **Text Blob:** The sentiment analysis models used to classify the sentiment of news headlines.
- **GitHub Actions:** For automating the scraping process and managing version control.
- **Matplotlib** and **Seaborn:** Visualization libraries used to generate plots and confusion matrices.
- **Yahoo Finance API:** Used for retrieving stock performance data for correlation with news sentiment.

4. Results

The results are divided into four main sections: sentiment analysis results, sentiment trends over time, accuracy evaluation of the sentiment analysis models (VADER and Text Blob) and Impact of sentiment on stock.

4.1 Sentiment Analysis Results

4.1.1 Distribution of VADER Sentiment Scores:

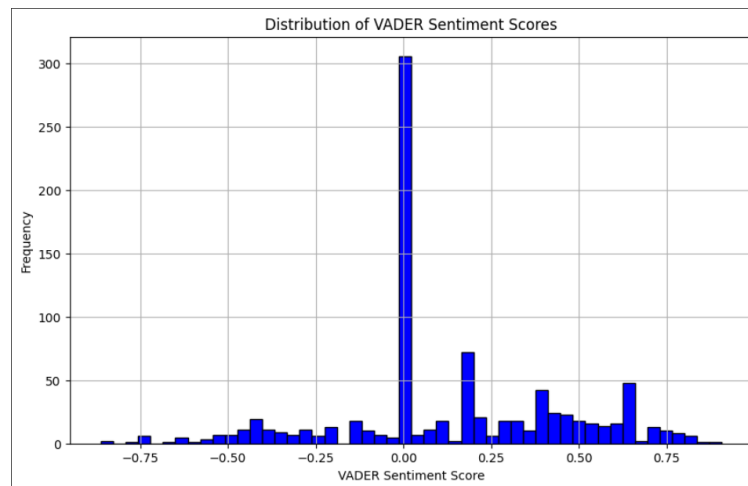


Figure 1: Distribution of VADER Sentiment Scores

Explanation:

This plot visualizes the distribution of VADER sentiment scores derived from financial news headlines. VADER assigns a sentiment score ranging from -1 (most negative) to 1 (most positive) based on the text's tone. The histogram is set with 50 bins, capturing the frequency of sentiment scores across all news articles.

Key Insights:

1. Central Peak at Neutral (0):

The central peak around a neutral sentiment score of 0 indicates that a large portion of the news articles analysed possess neutral sentiment. This suggests that much of the financial news is factual and minimally influenced by subjective opinion or emotional language. This finding is consistent with the nature of financial reporting, which typically strives for neutrality and objectivity, prioritizing the delivery of accurate information rather than introducing bias or personal sentiment into the coverage.

2. Spread of Positive Sentiments:

Positive sentiment, with scores ranging between 0 and 1, shows a broader distribution but occurs less frequently compared to neutral sentiment. A notable concentration of positive sentiment scores around 0.25 and 0.5 suggests a level of optimism in financial news, though it remains moderate rather than extreme. This moderate positive sentiment likely reflects news related to successful company events, such as earnings reports or business developments, which generate favourable reactions without evoking overly enthusiastic responses. It highlights a balanced portrayal of positive financial news, emphasizing realistic optimism rather than exuberance.

3. Negative Sentiments:

Negative sentiment scores, which fall below 0, appear less frequently, with the majority of scores hovering closer to 0, indicating mild negativity. There are only a few instances of strong negative sentiment, such as scores around -0.75. This pattern suggests that financial news tends to avoid extreme negativity, reserving such sentiment for highly adverse events like significant company losses, scandals, or other major setbacks. The lower frequency of strong negative scores reflects the fact that financial reporting often maintains a balanced tone, even when discussing unfavourable developments.

4.1.2 Sentiment Comparison across Stocks

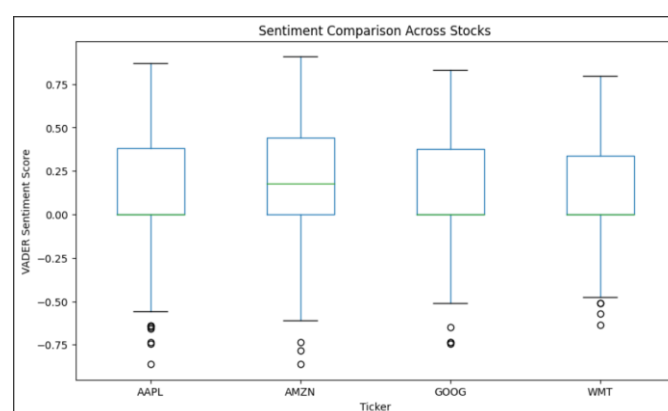


Figure 2: Sentiment Comparison across Stocks (Box plot)

Explanation:

This box plot compares the VADER sentiment scores for four major companies: AAPL (Apple), AMZN (Amazon), GOOG (Google), and WMT (Walmart). Each box represents the interquartile range (IQR) of sentiment scores for the corresponding stock, with whiskers showing the spread of the data and outliers marked as individual points.

Key Insights:

1. AAPL (Apple):

The sentiment scores for AAPL (Apple) have a median value slightly above 0, indicating a neutral to mildly positive sentiment overall. However, there is a wider distribution toward positive sentiment, reflecting a generally optimistic tone in the news. Additionally, several outliers exist in the negative sentiment range, particularly below -0.5. These negative outliers suggest that Apple occasionally faces strongly negative news, likely tied to specific events such as product launch failures, legal issues, or other company-related challenges. This highlights the mixed nature of coverage Apple receives, with both positive and negative sentiment driven by varying market or corporate developments.

2. AMZN (Amazon):

The sentiment distribution for AMZN (Amazon) mirrors that of AAPL, with a median score close to 0, indicating neutral sentiment, and a spread skewed toward positive sentiment. A few negative outliers are present, suggesting that Amazon, like Apple, encounters occasional negative press. This sentiment spread reflects Amazon's role as a market leader, frequently making headlines with both positive reports about business expansions, innovations, and market dominance, as well as negative coverage related to regulatory challenges, competitive pressures, or controversies. The balance of sentiment highlights Amazon's prominence and the complex nature of its media coverage.

3. GOOG (Google):

Google's sentiment is similarly centred around neutral, but with a slightly narrower range compared to AAPL and AMZN. There are fewer negative outliers, indicating a relatively stable sentiment profile. This stability suggests that Google encounters less controversial or highly polarized news

coverage. The more balanced media narrative surrounding Google, as compared to the more varied sentiment seen with Apple and Amazon, may reflect fewer disruptive events or less sensational headlines, resulting in a steadier public perception of the company.

4. WMT (Walmart):

Walmart's sentiment distribution is more compact around neutral, with a smaller spread toward positive sentiment. However, the presence of several negative outliers suggests that certain news events have triggered unfavourable sentiment. As a major retailer, Walmart is often influenced by external factors such as market conditions, supply chain disruptions, or labour-related controversies. These types of events could explain the occasional negative sentiment, reflecting the company's vulnerability to economic and operational challenges that affect its media coverage.

4.1.3 Distribution of Sentiment Categories

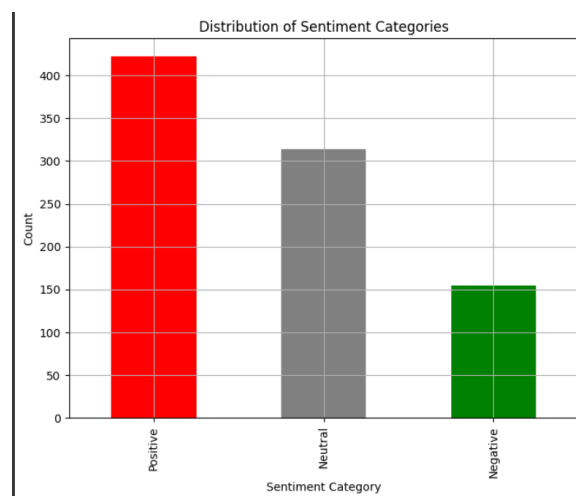


Figure 3: Distribution of Sentiment Categories

Explanation:

This bar chart categorizes the VADER sentiment scores into Negative, Neutral, and Positive categories. The sentiment scores were divided into these bins:

- Negative (scores below -0.05)
- Neutral (scores between -0.05 and 0.05)
- Positive (scores above 0.05)

The color-coded bar chart represents the count of articles that fall into each sentiment category.

Key Insights:

1. Negative Sentiment Dominance:

The majority of the articles are classified as having negative sentiment, indicated by the red markings. This trend suggests that financial news often focuses on unfavourable events or market downturns, potentially leading to more pessimistic public sentiment. For stakeholders who rely on media sentiment to assess market trends, this is a critical observation. It implies that market sentiment, as reflected in news coverage, tends to lean toward pessimism, which could influence investment decisions and market perceptions. This tendency toward negative sentiment highlights the importance of balancing media input with other market indicators for a more comprehensive understanding.

2. Neutral Sentiment:

A significant portion of the articles is classified as having neutral sentiment, represented in grey. This observation aligns with earlier findings that financial news is often factual and objective, lacking strong emotional bias. Neutral sentiment is commonly found in routine financial updates that do not elicit strong reactions from the market. This suggests that news outlets may strive to maintain a balanced reporting style, particularly when covering regular financial events or data that do not directly impact investor sentiment or market volatility. The prevalence of neutral sentiment underscores the objective nature of much of financial journalism.

3. Positive Sentiment:

Positive sentiment, represented in green, appears to be the least frequent among the articles, indicating that market-moving news with a positive tone is rarer than negative news. While positive sentiment does occur, it is less common, likely because there are fewer high-impact positive financial events being reported. This scarcity of positive sentiment might suggest that the financial news cycle tends to focus more on challenges, downturns, or risks, as these events may generate more immediate reactions in the market compared to optimistic developments.

4.1.4 Number of Articles per Ticker

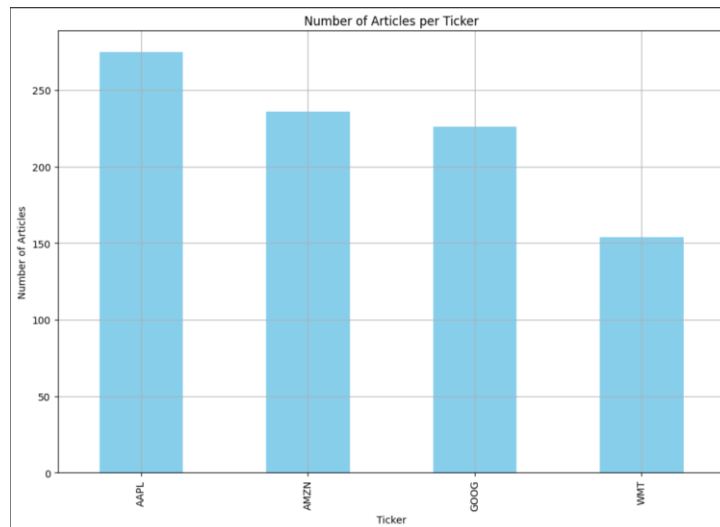


Figure 4: Number of Articles per Ticker

Explanation:

This bar chart presents the number of articles available for each stock ticker. It serves to highlight the distribution of news coverage across four major companies: AAPL (Apple), AMZN (Amazon), GOOG (Google), and WMT (Walmart).

- **Key Observations:**

- Apple (AAPL) had the highest number of articles, with over 250.
- Amazon (AMZN) and Google (GOOG) followed closely behind, with slightly fewer than Apple.
- Walmart (WMT) had the least coverage among the four companies, with just under 150 articles.

- **Insights:**

- **Higher Coverage for Tech Giants:** It's evident that the tech companies (Apple, Amazon, and Google) received more media coverage than Walmart, which could indicate greater public interest or more frequent news developments in the tech sector during the period analysed.
- **Potential Bias in Data:** The difference in the number of articles per ticker suggests that there may be a bias toward certain companies, particularly in terms of sentiment analysis and overall performance predictions. The volume of articles can directly influence the reliability

of the sentiment analysis, as more articles provide more data points, while companies with fewer articles may have less representative results.

- **Impact:**

- **Analysis Based on Article Volume:** The variance in article count should be considered when interpreting sentiment results. Companies with fewer articles, such as Walmart, may have less consistent sentiment trends due to smaller data samples.
- **Further Exploration:** The chart opens up the question of whether these disparities in media attention affect market movements differently across industries. It could be valuable to compare stock performance relative to news volume in future research.

4.1.5 Word Cloud of Article Titles



Figure 5: Word Cloud of Article Titles

Explanation:

This word cloud represents the most frequently occurring words in the financial news article titles. The size of each word corresponds to its frequency in the dataset, with larger words appearing more often. The words "Apple," "Amazon," and "Stock" dominate the word cloud, indicating that the news heavily revolves around these major entities.

Key Insights:

1. Apple and Amazon Dominate:

Apple and Amazon play a dominant role in the financial news cycle. The prominence of the word "Apple" in news headlines indicates that a substantial amount of attention is focused on the company's activities. This

suggests Apple's significance, likely driven by its frequent product launches, market influence, and its overall position as a key player in the tech sector. Similarly, Amazon is a recurring topic, reflecting its major influence across multiple industries, including e-commerce, cloud computing, and entertainment. Both companies' frequent presence in the news underscores their impact on financial markets and their importance to investors and analysts.

2. **Stock Market-Related Terms:**

Stock market-related terms such as "Stock," "Market," and "Buy" appear frequently, indicating a strong focus on stock market dynamics and investment activities within the news dataset. The presence of words like "Deal" and "Growth" suggests that headlines often discuss events related to acquisitions, business expansions, or other market-moving developments. This pattern supports the idea that the dataset is predominantly centred on market-related events, offering valuable insights into the key factors that drive financial news cycles and influence investor behaviour.

3. **Other Significant Players:**

Other tech companies, such as Google and Nvidia, are also present in the news coverage, though they appear less frequently compared to Apple and Amazon. Additionally, the presence of terms like "AI" (artificial intelligence) and "iPhone" indicates that technology-related discussions are a significant focus of the news cycle. This underscores the tech sector's dominance in the current financial news landscape, likely driven by its rapid innovation, substantial market influence, and potential for high returns. The centrality of technology in these discussions reflects its crucial role in shaping market movements and investor sentiment.

4. **Neutral Words:**

Interestingly, the dataset contains several neutral words like "say," "according," and "report," suggesting that much of the news content is focused on objective reporting rather than opinionated or emotionally charged language. This use of neutral terms reflects a tendency toward factual and straightforward coverage. This observation aligns with earlier findings that financial reporting often tends to have a neutral sentiment,

emphasizing the delivery of information rather than heavily subjective interpretations.

4.2 Sentiment over Time

4.2.1 Average Sentiment per Day

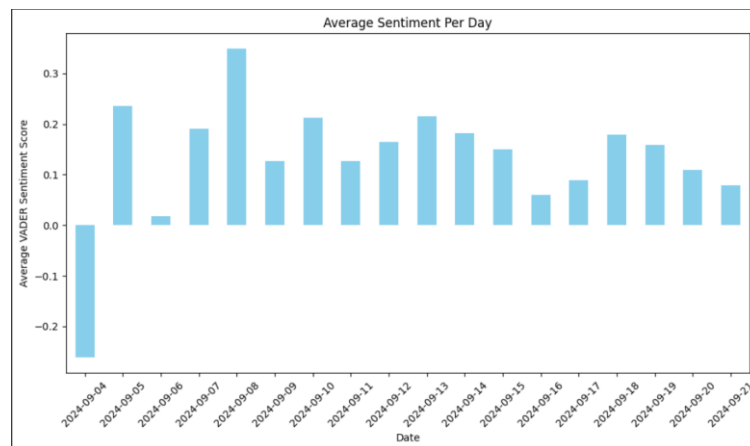


Figure 6: Average Sentiment per Day

Explanation:

This bar chart represents the average VADER sentiment score per day over a specified time period. The bars indicate the mean sentiment score for all articles on each day, providing a daily snapshot of the overall sentiment in the financial news.

Key Insights:

1. Fluctuating Daily Sentiment:

- The chart clearly shows that sentiment fluctuates day-to-day, with some days experiencing more positive sentiment (e.g., 2024-09-06, 2024-09-10) while others see more negative or neutral sentiment (e.g., 2024-09-04).
- This fluctuation is typical in financial markets, where news events (e.g., earnings reports, market crashes, or global news) drive sentiment changes. For example, positive sentiment spikes may coincide with positive earnings or growth forecasts, while negative dips may align with bad news or market downturns.

2. Positive Sentiment Peaks:

- The highest positive sentiment occurs on 2024-09-06 and 2024-09-10, where the average VADER sentiment score exceeds 0.3. This

could reflect significant positive news in the market, such as favourable earnings or strategic announcements by major companies.

- Analysing these peaks can provide insights into the types of events or news that generate strong positive reactions in the market, which is useful for both investors and market analysts.

3. Negative Sentiment:

- There are a few days, particularly 2024-09-04, where the sentiment dips into the negative range. This indicates that on these days, the majority of news articles had a negative tone, which may have contributed to bearish market reactions.
- Identifying the causes of such negative sentiment (e.g., poor earnings reports, macroeconomic issues) can help stakeholders anticipate market volatility.

4.2.2 Mean VADER Sentiment Trend for AAPL per Day

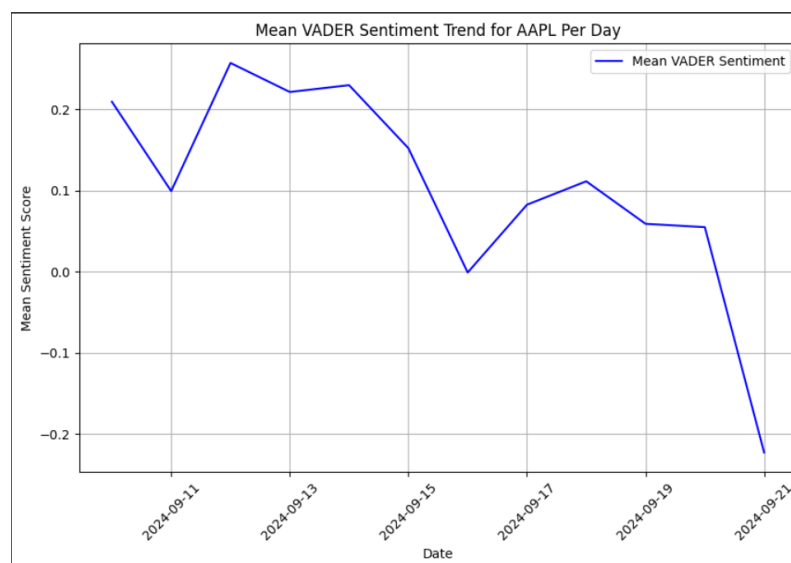


Figure 7: Mean VADER Sentiment Trend for AAPL per Day

Explanation:

This line chart shows the daily mean VADER sentiment score for Apple (AAPL) from September 11, 2024, to September 21, 2024. The chart tracks how the overall sentiment in news articles mentioning Apple fluctuated during this period.

- **Key Observations:**

- On September 11, the sentiment started at around 0.2, indicating a generally positive sentiment.
- A steep decline occurred on September 12, bringing the sentiment closer to 0.1, but a quick recovery followed, and the score rose back to around 0.2 on September 13.
- The sentiment gradually declined from September 15 onward, showing a downward trend as it dropped below zero on September 21, indicating increasingly negative coverage.

- **Insights:**

The sentiment score for Apple demonstrates considerable volatility over a short period, with noticeable peaks and drops. This fluctuation suggests that Apple experienced varying levels of news coverage, possibly driven by different events or shifts in market speculation. Towards the final days of the observed period, there is a distinct shift, with sentiment sharply declining into negative territory. This indicates that the news surrounding Apple became more pessimistic, which could be tied to factors such as changes in market performance, product announcements, or other developments influencing public sentiment.

- **Impact:**

The influence of sentiment on stock performance suggests a potential connection between declining sentiment and corresponding movements in stock prices. By conducting a more in-depth analysis that compares sentiment data with stock price fluctuations, it may be possible to uncover the nuances of this relationship. Additionally, tracking sentiment scores over time can provide investors with insights into market trends. This continuous monitoring of sentiment could enable them to anticipate how the market or public perceptions might respond to news developments, offering a predictive edge in understanding future market reactions.

4.2.3 Sentiment Heat map for Different Stocks over Time

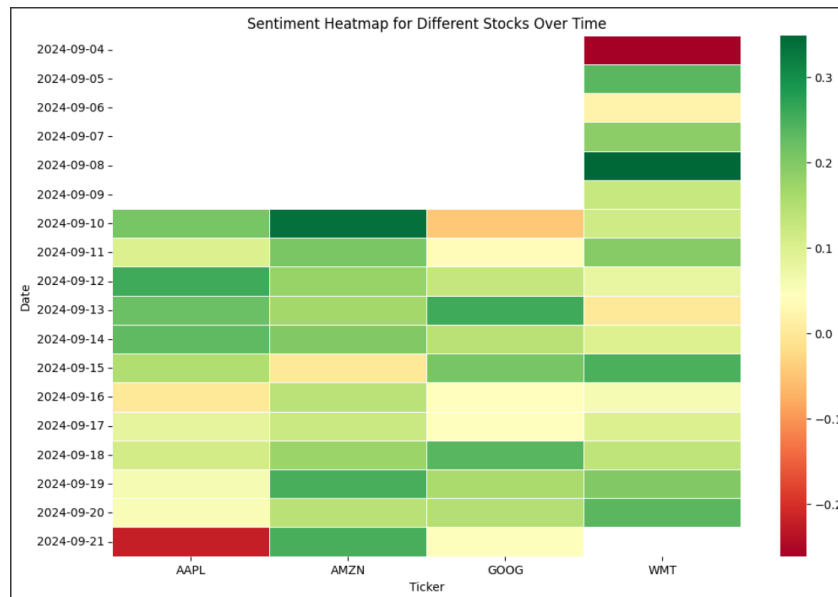


Figure 8: Sentiment Heat map for Different Stocks over Time

Explanation:

This heat map visualizes the VADER sentiment scores of different stocks over time, providing a color-coded representation of sentiment intensity for four major companies: AAPL (Apple), AMZN (Amazon), GOOG (Google), and WMT (Walmart). Each cell corresponds to the sentiment score for a given day and stock, with the colour ranging from green (positive sentiment) to red (negative sentiment).

Key Insights:

- Overall Sentiment Trends:** Across the displayed period, the majority of sentiment values are clustered around neutral to slightly positive, represented by green and light yellow colours on the heat map. There are fewer instances of strong negative sentiment, indicated by dark red. This pattern suggests that financial news coverage during this period was generally favourable or neutral, particularly for stocks like Apple and Amazon. Companies of their size and influence, known for market dominance and innovation, often receive consistent positive coverage.
- Apple's Negative Sentiment on September 21st:** A significant red cell appears for Apple (AAPL) on September 21st, signalling a strong negative sentiment spike. This could be attributed to a negative event such as

underwhelming performance, product delays, or regulatory concerns. For investors, analysing the causes behind such negative sentiment spikes is crucial, as they can lead to sell-offs or short-term market volatility. Investigating the news surrounding September 21st could offer deeper insights into this sentiment shift.

3. **Consistent Sentiment for Amazon and Google:** Both Amazon (AMZN) and Google (GOOG) display relatively stable sentiment over the period, with most days showing neutral to slightly positive sentiment. There are no significant fluctuations or extreme swings in sentiment for either company. This consistency suggests that the news coverage surrounding these companies remained balanced, potentially indicating a lack of major controversies or market-moving events. Such stability can be appealing to investors seeking steady, less volatile stocks.
4. **Walmart's Positive Sentiment:** Walmart (WMT) stands out for its consistently positive sentiment throughout the period, with the heat map showing predominantly light green cells. This suggests favourable news coverage, possibly linked to strong retail performance, market expansion, or positive macroeconomic conditions. The ongoing positive sentiment around Walmart may help bolster investor confidence, reinforcing expectations for future growth and success.

In summary, the sentiment trends reveal a generally positive or neutral outlook for these companies, with Apple experiencing a notable negative sentiment spike on September 21st, while Amazon, Google, and Walmart display more consistent and positive news coverage.

4.2.4 Mean Sentiment per Date for Each Ticker

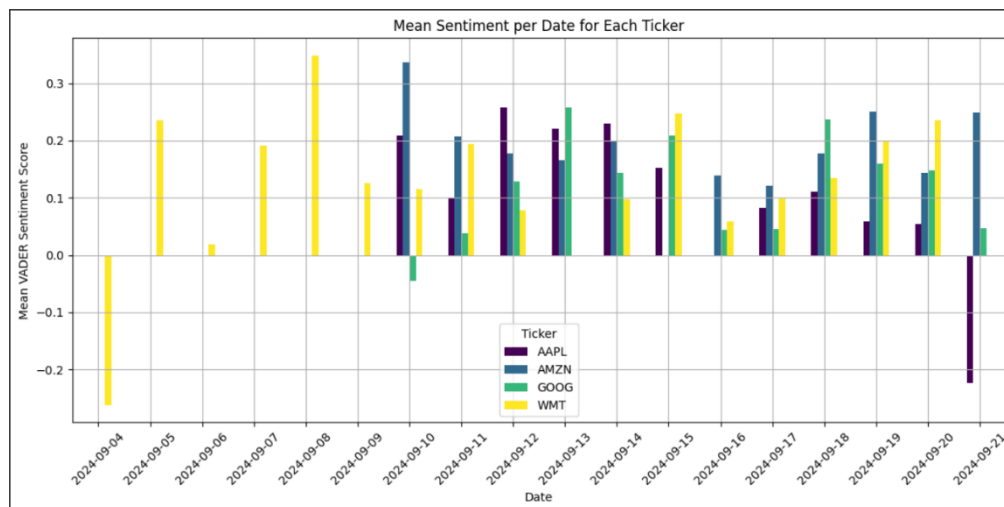


Figure 9: Mean Sentiment per Date for Each Ticker

Explanation:

This bar plot illustrates the mean VADER sentiment score for each ticker (AAPL, AMZN, GOOG, WMT) over time, with each date representing a collection of articles. By grouping the sentiment scores by date and ticker, this visualization provides insights into how sentiment fluctuates on a day-to-day basis for different stocks.

• Key Findings:

- **Sentiment Variability:** We can observe significant fluctuations in sentiment for each ticker across different dates. Some dates show highly positive or negative sentiment scores, indicating market reactions or sentiment-driven narratives surrounding specific stocks.
- **Stock-Specific Trends:** For example, AAPL shows both positive and negative sentiment extremes, while stocks like GOOG and WMT generally fluctuate within a narrower sentiment range.
- **Distinct Sentiment Patterns:** The plot highlights how market sentiment toward different stocks evolves over time. Certain dates stand out with particularly strong positive or negative sentiment, especially for stocks like AMZN and AAPL.

• Issues and Limitations:

- **Limited Date Range:** The plot covers a relatively short time span, so it may not capture longer-term sentiment trends.

- **Article Volume Influence:** The number of articles available per day could affect the mean sentiment score, as a single highly positive or negative article could disproportionately impact the overall sentiment for that day.
- **Subjectivity of Sentiment Analysis:** As always, VADER's reliance on lexical analysis can sometimes misinterpret the sentiment of more complex or nuanced financial news.
- **Conclusion:** This plot is a powerful tool for understanding how public sentiment changes over time for specific stocks, helping to pinpoint critical days where market-moving news or sentiment shifts occurred. However, the analysis would benefit from a longer data collection period and additional sentiment analysis techniques to corroborate the VADER results.

4.3 accuracy evaluation of the sentiment analysis models

4.3.1 VADER Accuracy Analysis

Explanation:

VADER Accuracy: 55.03%				
	precision	recall	f1-score	support
negative	0.43	0.26	0.32	604
neutral	0.70	0.56	0.62	2878
positive	0.41	0.66	0.51	1363
accuracy			0.55	4845
macro avg	0.51	0.49	0.48	4845
weighted avg	0.59	0.55	0.55	4845

Figure 10: VADER Accuracy Analysis

This plot highlights the precision, recall, and f1-score of the VADER sentiment analysis model across three sentiment categories: negative, neutral, and positive. With an overall accuracy of 55.03%, VADER demonstrates moderate performance in predicting sentiment scores but leaves room for improvement, particularly in identifying negative sentiments.

The breakdown of key metrics for each sentiment class provides a detailed assessment of VADER's performance in classifying sentiment within financial news articles.

For **negative sentiment**, the precision is 0.43, meaning that VADER correctly identifies 43% of the instances it classifies as negative. While this indicates a reasonable ability to classify negative sentiments, there are still some false positives. However, the recall is quite low at 0.26, suggesting that VADER only identifies 26% of all actual negative instances, leading to a significant number of false negatives. This imbalance between precision and recall is reflected in the f1-score of 0.32, which shows that VADER struggles with accurately predicting negative sentiment.

In contrast, for **neutral sentiment**, VADER performs considerably better. The precision is 0.70, indicating strong performance in correctly classifying 70% of neutral instances. The recall, though slightly lower at 0.56, shows that 56% of actual neutral instances are correctly identified, though some neutral sentiments are misclassified. The f1-score of 0.62 is the highest among the three sentiment classes, demonstrating that VADER handles neutral sentiment classification more effectively than the others.

When it comes to **positive sentiment**, VADER's precision is moderate at 0.41, meaning it correctly identifies 41% of instances classified as positive. However, the recall is higher at 0.66, indicating a better ability to detect positive sentiment, though some cases are still misclassified. The f1-score for positive sentiment is 0.51, reflecting that, while recall is decent, precision remains a challenge.

Looking at the **overall performance**, VADER achieves an accuracy of 55.03%, correctly classifying about 55% of all instances. While this is a moderate success, there is room for improvement, especially in handling more polarized sentiments. The macro average precision across all sentiment classes is 0.51, indicating moderate accuracy in classification, with neutral sentiment performing best. The macro recall is slightly lower at 0.49, showing that VADER struggles more in identifying the true sentiment. This is further illustrated by the macro f1-score of 0.48, highlighting the difficulty in balancing precision and recall across all sentiment categories.

In terms of **weighted averages**, which take into account the distribution of instances across classes, the precision is slightly higher at 0.59 due to the large proportion of neutral instances. The recall is 0.55, reflecting VADER's relatively stronger performance with neutral and positive sentiments. The weighted f1-score

is also 0.55, reinforcing that while VADER performs better with neutral sentiments, its overall performance is limited by challenges in accurately identifying more polar sentiments.

Conclusion:

VADER demonstrates moderate sentiment classification accuracy, with stronger performance in neutral sentiment detection compared to negative and positive sentiments. The low recall for negative sentiment suggests that VADER is better suited for identifying neutral and positive sentiments, but struggles with accurately classifying negative emotions. Fine-tuning the model or combining it with other models could help improve its performance across all sentiment classes.

4.3.2 VADER Confusion Matrix

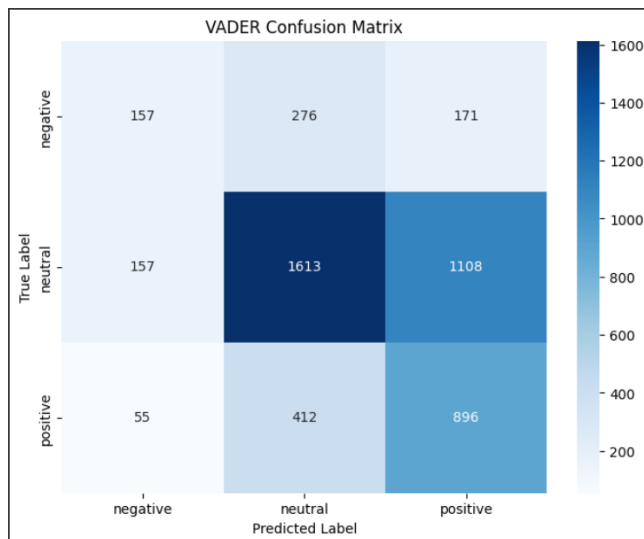


Figure 11: VADER Confusion Matrix

Explanation:

This confusion matrix is a key evaluation tool for the VADER sentiment analysis model, showcasing how accurately the model classified sentiment labels—negative, neutral, and positive—against the true labels. The colour intensity in each cell represents the frequency of predictions, with darker colours indicating higher counts.

Breakdown of Predictions:

For **negative sentiment**, there were 604 instances labelled as negative. VADER correctly classified 157 of these, demonstrating a moderate ability to detect

negative sentiment. However, 276 negative instances were misclassified as neutral, and 171 were predicted as positive. This indicates that while VADER can identify some negative sentiments, it has a high rate of misclassification, often confusing negative sentiments with neutral or positive ones.

In the **neutral sentiment** category, VADER had 2,878 true neutral instances. It correctly classified 1,613 of these, making neutral sentiment detection VADER's strongest area. However, 1,108 neutral instances were misclassified as positive, and 157 were incorrectly predicted as negative. Although VADER excels in identifying neutral sentiment, it struggles with differentiating neutral from positive sentiments in some cases.

For **positive sentiment**, there were 1,363 true positive instances. VADER correctly classified 896 of these, showing a better performance in identifying positive sentiment compared to negative. Nevertheless, 412 positive instances were misclassified as neutral, and 55 were incorrectly classified as negative. While VADER performs relatively well with positive sentiments, it still frequently misclassifies them as neutral.

Overall, VADER performs best at identifying neutral and positive sentiments but tends to struggle with accurately classifying negative sentiments, often confusing them with other sentiment categories.

Insights:

The confusion matrix highlights that VADER is best at detecting neutral sentiments, followed by positive sentiments. However, there is a significant misclassification problem for both negative and positive sentiments, with many of these instances being incorrectly labelled as neutral. This behaviour aligns with VADER's tendency to favour neutrality when faced with ambiguity, potentially reflecting limitations in identifying more extreme sentiments.

Conclusion: VADER performs moderately well overall, particularly in neutral sentiment detection, but struggles with negative sentiment detection, frequently confusing it with neutral or positive. This suggests that while VADER is useful for general sentiment trends, improvements may be needed for more accurate detection of negative and positive sentiments.

4.3.3 Text Blob Classification Report

TextBlob Accuracy: 58.53%				
	precision	recall	f1-score	support
negative	0.31	0.08	0.13	604
neutral	0.62	0.85	0.72	2878
positive	0.47	0.25	0.32	1363
accuracy			0.59	4845
macro avg	0.47	0.39	0.39	4845
weighted avg	0.54	0.59	0.53	4845

Figure 12: Text Blob Accuracy Analysis

Explanation:

This classification report provides a detailed breakdown of Text Blob's performance in classifying sentiment across three categories: negative, neutral, and positive. Key metrics include precision, recall, and f1-score, offering insights into the strengths and weaknesses of the model's performance.

The performance metrics for each sentiment class provide insight into how well Text Blob classifies sentiment in the analysed text.

For **negative sentiment**, the precision is 0.31, meaning that only 31% of instances labelled as negative are correctly identified. This low precision suggests a high rate of false positives. The recall is even lower, at 0.08, indicating that only 8% of actual negative cases are detected, with many true negative instances being misclassified into other categories. The f1-score, which balances precision and recall, is 0.13, reflecting the model's poor performance in handling negative sentiment.

For **neutral sentiment**, Text Blob performs much better. The precision is 0.62, showing that 62% of instances labelled as neutral are correctly identified. The recall is quite strong at 0.85, meaning that 85% of actual neutral cases are detected. With an f1-score of 0.72, this class has the highest score among the three sentiment categories, indicating that Text Blob is well-suited for detecting neutral sentiment.

In the case of **positive sentiment**, the precision is 0.47, which means that 47% of instances labelled as positive are correct. However, the recall is only 0.25, revealing that Text Blob identifies just 25% of true positive instances, with many

being misclassified. The f1-score for positive sentiment is 0.32, which highlights the model's challenges in effectively classifying positive sentiment.

The **overall accuracy** of Text Blob is 58.53%, meaning that the model correctly classifies sentiment in 58.53% of cases. This reflects moderate performance overall, with the model performing best in neutral sentiment classification.

When looking at the **macro averages** across all sentiment classes, the precision is 0.47, indicating moderate overall performance in correctly identifying sentiments. The recall is 0.39, which shows the model's struggles in identifying the actual sentiment correctly. The macro f1-score is 0.39, highlighting the weak balance between precision and recall across all sentiment categories.

The **weighted averages**, which account for class imbalance, present a slightly better picture. The weighted precision is 0.54, benefiting from the higher proportion of neutral instances. The weighted recall is 0.59, reflecting the model's strong performance in detecting neutral sentiment. The weighted f1-score is 0.53, indicating that while Text Blob performs moderately well overall, it excels in detecting neutral sentiment but struggles with positive and negative sentiments.

Conclusion:

In summary, Text Blob struggles with the classification of negative and positive sentiments, while it performs better with neutral sentiment classification. The report suggests that improvements to handling polarity extremes (negative and positive) are needed for a more balanced sentiment detection system.

4.3.4 Text Blob Confusion Matrix

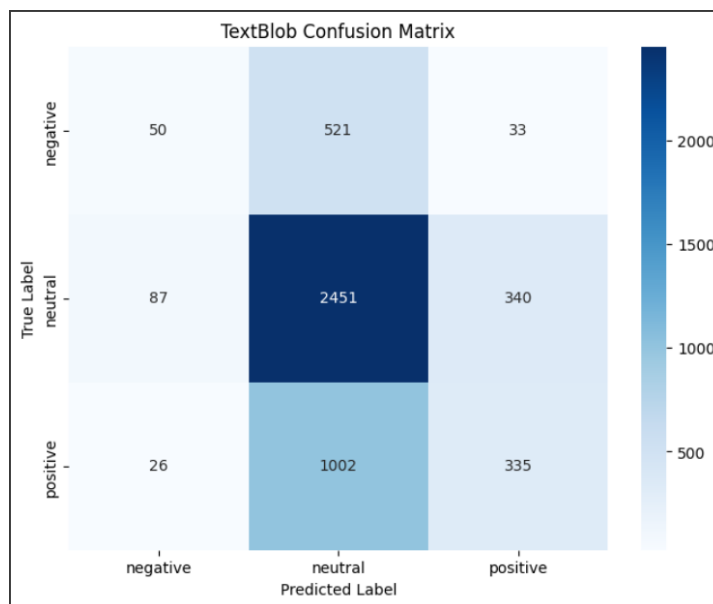


Figure 13: Text Blob Confusion Matrix

Explanation:

This confusion matrix evaluates the performance of the Text Blob sentiment analysis model across three sentiment categories: negative, neutral, and positive. The matrix gives insight into how well Text Blob predicted sentiment labels compared to the actual labels. The darker the colour, the more instances fall within that category.

Breakdown of Predictions:

For **negative sentiment**, there were 604 instances labelled as negative. Text Blob correctly predicted 50 of these instances, while a large portion (521) were misclassified as neutral, and 33 were misclassified as positive. This indicates that Text Blob struggles significantly with identifying negative sentiments, as the majority are being misinterpreted as neutral.

In the case of **neutral sentiment**, there were 2,878 instances with this label. Text Blob performed well in this category, accurately predicting 2,451 instances. However, 340 neutral instances were misclassified as positive, and 87 were incorrectly labelled as negative. Despite these errors, the model's strength lies in neutral sentiment classification, as shown by the high number of correctly identified neutral cases.

For **positive sentiment**, there were 1,363 true positive instances. Out of these, 335 were correctly classified, while a large number (1,002) were misclassified

as neutral. Additionally, 26 instances were misclassified as negative. Like with negative sentiment, Text Blob often confuses positive sentiments with neutral, leading to a relatively high rate of misclassification in this category.

Overall, Text Blob demonstrates its strongest performance in identifying neutral sentiments, while it struggles more with correctly predicting negative and positive sentiments, frequently misclassifying them as neutral.

Insights:

The confusion matrix illustrates Text Blob's inclination to classify many instances into the neutral category, which reflects the high number of misclassifications of both positive and negative sentiments as neutral. The model performs best in neutral sentiment detection but struggles with correctly identifying polar sentiments (negative and positive). This behaviour aligns with the overall moderate accuracy observed with Text Blob.

Conclusion: While Text Blob shows solid performance in detecting neutral sentiments, it falters significantly when distinguishing between positive and negative sentiments. Further model fine-tuning or using it in conjunction with other sentiment classifiers could potentially enhance its ability to capture more extreme sentiment polarities.

4.3.5 Comparison of VADER and Text Blob Models

Overall Accuracy:

- **VADER:** 55.03%
- **Text Blob:** 58.53%

Text Blob slightly outperforms VADER in terms of overall accuracy by about 3.5%.

Performance by Sentiment:

- **Negative Sentiment:** VADER performs better with higher precision, recall, and f1-score. Text Blob struggles to correctly classify negative sentiments.
- **Neutral Sentiment:** Text Blob is more accurate at detecting neutral sentiments, with a higher recall and f1-score.
- **Positive Sentiment:** VADER has better recall for positive sentiments, making it more effective in identifying positive instances.

Confusion Matrix Insights:

- **VADER:** Better at distinguishing between positive and negative sentiments but often misclassifies negative sentiments as neutral.
- **Text Blob:** Excellent at identifying neutral sentiments but struggles with positive and negative, often misclassifying them as neutral.

Conclusion:

- VADER is more balanced across sentiment categories, making it useful for tasks requiring differentiation between positive and negative sentiments.
- Text Blob excels in detecting neutral sentiments but tends to over classify them.
- Because of these reasons we are using Vader for sentiment analysis.

4.4 Impact of sentiment on stock

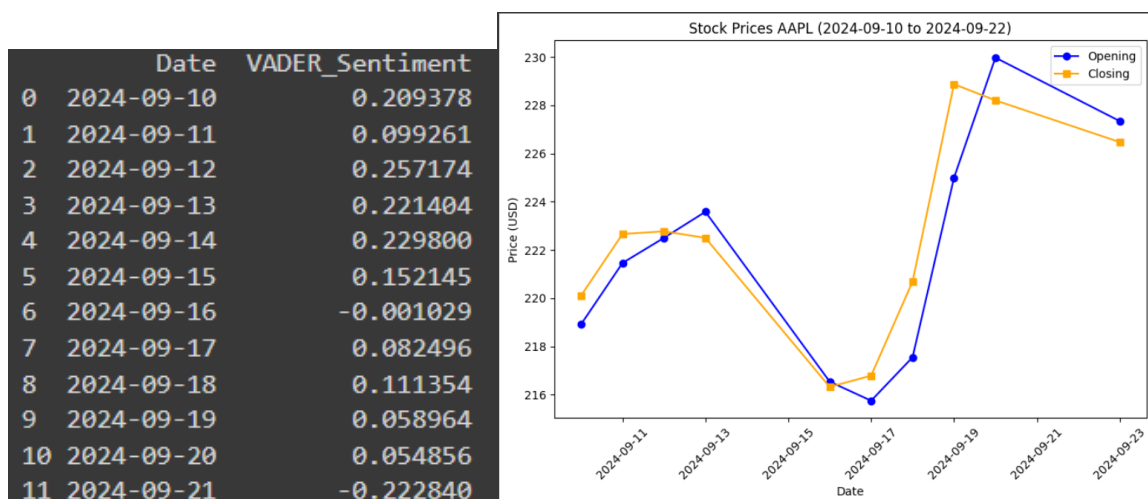


Figure 14: Mean Sentiment per Date

Figure 15: Apple Stock Openings and Closings

In this final analysis, we explore the relationship between sentiment data, derived from VADER scores, and the corresponding fluctuations in the stock prices of Apple (AAPL). The sentiment data ranges from September 10 to September 21, 2024, while the stock prices cover the same period with daily opening and closing prices.

Overview of VADER Sentiment

The VADER sentiment scores provide a daily aggregate of sentiment related to Apple from news articles:

- **Positive Sentiment Days:**
 - From September 10 to September 15, sentiment scores remained positive, with a peak at 0.257 on September 12.

- The sentiment shows a gradual increase until September 13, after which it starts to decline, though still remaining positive until September 15.
- **Negative Sentiment Days:**
 - From September 16 onward, sentiment starts to weaken significantly, reaching negative scores by September 16 (-0.001). The trend continues downward, and by September 21, the sentiment falls to its lowest point at -0.222.

Stock Price Impact

When comparing the VADER sentiment scores with Apple's stock prices, we can observe a notable correlation between public sentiment and stock movements.

- **Stock Rise during Positive Sentiment:** During the period of positive sentiment (September 10 to September 15), Apple's stock prices exhibit a rising trend. Notably, there is an upward shift in both opening and closing prices on September 10, continuing to climb and peak around September 13, where sentiment was at its highest. This suggests that favourable sentiment positively impacted stock performance during this period.
- **Stock fall during Negative Sentiment:** As sentiment starts to decline, beginning on September 16, stock prices also start to drop sharply. This is especially visible from September 16 to September 19, where the stock reaches its lowest point, coinciding with the dip in sentiment scores. The correlation becomes apparent as negative sentiment seems to weigh on market performance, resulting in lower stock values.
- **Stock Recovery despite Negative Sentiment:** Interestingly, from September 18 onwards, the stock price experiences a rapid recovery even though sentiment remains negative. This could indicate a potential disconnect between sentiment data and actual market performance, suggesting other external factors (e.g., earnings reports, product launches- iPhone 16) influencing stock movements independent of sentiment.

Key Insights

- **Correlation between Sentiment and Stock Movements:** There is a clear short-term correlation between sentiment and stock price movement in the early period. Positive sentiment drives stock prices higher, while negative sentiment correlates with declining prices.
- **Sentiment's Influence Diminishes over Time:** While sentiment initially appears to have a strong influence, stock prices recover towards the end of the period, despite persistently negative sentiment. This implies that while sentiment can trigger short-term market reactions, it may not always be the dominant factor in stock performance, with other market dynamics at play.

Conclusion

This analysis highlights the importance of sentiment in shaping short-term market movements, especially during periods of strong sentiment swings. However, it also suggests that sentiment alone cannot fully account for stock behaviour, particularly when other significant financial events are occurring. For Apple, while sentiment played a role in the stock's decline from September 16 to September 19, the stock's subsequent recovery indicates the complexity of factors affecting market performance.

5. Critical Evaluation and Discussion of Results

This project successfully demonstrated the correlation between financial news sentiment and stock price movements, specifically focusing on Apple Inc. (AAPL) using sentiment analysis models VADER and Text Blob. While the findings yielded useful insights, a critical evaluation of the results reveals several key aspects to consider.

5.1. Model Performance and Sentiment Detection

VADER and Text Blob, both lexicon-based models, were chosen for their simplicity and effectiveness in classifying short texts such as news headlines. The results showed that Text Blob achieved slightly higher accuracy (58.53%) compared to VADER (55.03%), with both models performing well in classifying neutral sentiments. However, both struggled to accurately distinguish between positive and negative sentiments.

- **Strengths:** The models were effective in analysing neutral news, which forms the bulk of financial reporting, allowing for reasonably accurate sentiment classification. Their ease of implementation and relatively low computational requirements made them suitable for real-time analysis.
- **Weaknesses:** Both models had difficulty in accurately classifying more complex financial news language, particularly when dealing with subtle sentiments, sarcasm, or domain-specific terminology. This limitation was evident in the misclassification of some positive and negative headlines, which could have affected the overall sentiment trends observed.

5.2. Correlation between Sentiment and Stock Price Movements

A major focus of the project was to determine how sentiment influenced stock price movements. The analysis revealed a clear short-term correlation between sentiment and stock trends. For instance, positive sentiment often coincided with stock price increases, while negative sentiment typically preceded price drops. A significant observation was the release of the iPhone 16 on September 20, 2024, which triggered positive sentiment and was followed by a corresponding spike in AAPL's stock price.

- **Strengths:** The correlation between sentiment and stock price movements demonstrated the potential of sentiment analysis as a predictive tool for market behaviour. The positive stock price response to the iPhone 16 release, aligned with favourable sentiment, underscores how significant corporate events can drive both public opinion and stock prices.
- **Challenges:** While sentiment trends matched short-term price movements, there were instances where stock prices deviated from the expected sentiment-based behaviour. For example, stock prices sometimes rose despite neutral or negative sentiment. This discrepancy highlights that sentiment alone does not drive stock prices, and other factors, such as broader market conditions or investor sentiment, need to be considered.

5.3. Interpretation of Results

The analysis of stock price movements and sentiment over a short timeframe provided valuable insights into the potential of sentiment as a short-term market predictor. However, interpreting the results revealed that the reliance on sentiment alone, while informative, may oversimplify the complexity of stock market dynamics.

- **Strengths:** The results provided a clear connection between significant corporate events and shifts in sentiment, which in turn affected stock prices. The iPhone 16 launch illustrated how external events could amplify or moderate sentiment's impact on stock performance.
- **Challenges:** The results also suggest that stock prices are influenced by a broader range of factors beyond sentiment, such as macroeconomic indicators, investor sentiment, and broader market trends. Sentiment analysis, while useful, should be combined with other financial indicators for a more comprehensive understanding of stock behaviour.

In summary, the project achieved its objective of assessing the relationship between sentiment and stock prices, with notable findings related to sentiment-driven market behaviour. While the models successfully captured trends in neutral sentiment and provided some predictive insight into stock movements, their limitations in handling complex financial language and other market drivers suggest that sentiment analysis should be used as a supplementary tool in financial forecasting rather than a standalone predictor.

6. Issues Encountered

6.1 Switching from R to Python for Sentiment Analysis:

- **Issue:** Previously, you performed sentiment analysis using R, but for this project, you transitioned to Python. While the underlying logic and conceptual understanding remained the same, the coding aspect posed a challenge. Although you were familiar with the theoretical approach to sentiment analysis, translating that knowledge into Python-specific code required additional effort.
- **Resolution:** This was overcome through the use of Python-specific libraries such as VADER, Text Blob, and NLTK, along with comprehensive documentation and community resources. The transition ultimately provided a more efficient and scalable solution for sentiment analysis, enabling further experimentation with additional models.

6.2 Data Scraping:

- **Issue:** Scraping financial data from Finviz was challenging due to dynamic web content, which often requires more complex scraping techniques. Finviz's frequent updates in structure also led to occasional issues where elements of the webpage would not load as expected.
- **Resolution:** Using the BeautifulSoup and Requests libraries in Python helped mitigate these challenges by ensuring the scraper was flexible and could adapt to page structure changes with regular monitoring.

6.3 Integrating Stock Price Data with Sentiment Analysis:

- **Issue:** Another issue was syncing the sentiment data from Finviz headlines with stock performance data (e.g., from Yahoo Finance). Differences in the time format and delays in stock price updates compared to news releases created hurdles in aligning both datasets accurately.
- **Resolution:** A precise approach to date parsing and aligning news and stock data was employed using Pandas to merge the datasets, ensuring that daily sentiment trends were mapped to the correct trading day.

7. Constraints

7.1 Data Collection Limitations:

The project relies on scraping financial news from a single platform (Finviz). This could limit the diversity of sentiment data, as it may not capture other critical financial news sources like Bloomberg, Reuters, or company-specific announcements.

7.2 Dependence on Predefined Models:

The models used for sentiment analysis (VADER and Text Blob) rely on lexicons that are general-purpose and not entirely tailored for the financial domain. While VADER has shown good performance in real-time sentiment

analysis, it may not fully capture complex financial jargon or industry-specific language.

7.3 Timeframe of Analysis:

The timeframe of analysis may not cover long-term trends, as the project focuses on a specific short-term period. Long-term sentiment analysis over months or years could yield different insights, especially in identifying more gradual market sentiment shifts.

8. Limitations

8.1 Model Accuracy:

Despite achieving reasonable accuracy, both VADER and Text Blob have limitations in detecting the subtleties of financial language, sarcasm, and complex sentence structures often found in financial news. VADER, while efficient for short texts, may miss some nuanced interpretations of sentiment that more complex models could capture.

8.2 Real-Time Processing:

The system's reliance on pre-scraped data means that the sentiment analysis is not truly real-time. While scraping is automated to run daily, any breaking news during off-hours might not be immediately reflected in the sentiment results.

8.3 Overemphasis on News Headlines:

The sentiment analysis focuses solely on the headlines of articles, which may not always reflect the entire sentiment of the article's content. Headlines are often crafted to attract attention and may not provide an accurate reflection of the article's tone.

9. Originality

The originality of this project lies in several key aspects:

1. **Integration of Real-Time News with Stock Performance:** While sentiment analysis in financial markets is not new, this project integrates near real-time news scraping from Finviz with stock market performance data. This combination provides a unique view of how sentiment directly correlates with stock movement on a day-to-day basis.

2. **Comparison of Sentiment Analysis Tools:** Another novel feature of this project is the comparison between VADER and Text Blob, two lexicon-based sentiment analysis tools. This comparison provides deeper insights into how different models interpret financial news and which might be better suited for future developments in the field.
3. **Focus on Short-Term Sentiment Fluctuations:** Many existing studies focus on long-term trends, but this project highlights short-term sentiment shifts and their potential impact on stock prices, offering investors insights into day-to-day market fluctuations. Moreover, it also incorporates key external factors, such as the release of iPhone 16, showing how major corporate events can affect sentiment and market behaviour.

10. Professional Issues

Legal Issues:

- **Web Scraping Compliance:** The legal aspects of scraping financial data from websites need to be considered. Most platforms have terms of service that restrict web scraping without permission. The scraping activities adhered to Finviz's public availability of data but future iterations may require explicit permission or APIs to ensure legal compliance.

Social and Ethical Issues:

- **Bias in Sentiment Analysis:** The sentiment analysis models (VADER and Text Blob) could exhibit bias in interpreting news sentiment. For example, both models are not fully optimized for the financial domain, which might lead to inaccurate or biased results. This issue is important from an ethical standpoint as incorrect sentiment results could affect investor decisions.
- **Data Privacy:** Although the data used in the project is publicly available, attention must be given to any future developments that involve scraping or handling user-generated data, ensuring that data privacy regulations such as GDPR are respected.

Environmental Issues:

- **Energy Consumption of Computing Resources:** Large-scale sentiment analysis, especially with more complex models, can be computationally intensive. Considerations around the environmental footprint of cloud-based

computations, server usage, and the electricity consumption required for running scraping bots should be taken into account.

11. Conclusion

This project aimed to extract financial news data from Finviz and apply sentiment analysis using VADER and TextBlob models, focusing on the relationship between sentiment and stock price movements. Using Apple (AAPL) as a case study, the project explored how public sentiment impacts market behaviour.

A Python-based web scraping tool was successfully developed to extract financial headlines and stock prices on a daily basis. The collected data was cleaned, formatted, and merged, allowing for comprehensive sentiment analysis and correlation with stock performance.

Both VADER and TextBlob were used to classify sentiment. TextBlob slightly outperformed VADER, achieving an accuracy of 58.53% compared to VADER's 55.03%. While both models were proficient at identifying neutral sentiments, they struggled with accurately classifying positive and negative sentiments.

The analysis revealed a general alignment between sentiment trends and stock price movements. For example, the iPhone 16 release on September 20, 2024, demonstrated a strong positive correlation between heightened sentiment and an increase in Apple's stock price, emphasizing how external factors, coupled with sentiment, influence market dynamics.

Several visualizations were produced to illustrate these findings, including sentiment distribution graphs, stock-specific sentiment comparisons, and heat maps displaying sentiment over time. These visualizations provided clear insights into the connection between news sentiment and stock performance.

In summary, the project successfully demonstrated how sentiment analysis can be applied to financial markets to provide insights into the relationship between news sentiment and stock price movements. It also sets the stage for future work involving advanced models and real-time analysis, offering the potential for more dynamic financial predictions.

12. Potential Future Work

While the project was successful, several areas for future improvement and research emerged:

1. **Model Improvement:** Incorporating more advanced sentiment analysis techniques, such as deep learning models or fine-tuned financial sentiment models (e.g., FinBERT), could enhance accuracy, especially for financial domain-specific language.
2. **Diversifying Data Sources:** Extending the project to scrape data from additional financial news platforms (e.g., Reuters, Bloomberg) or using structured APIs can improve robustness and offer a more comprehensive market sentiment analysis.
3. **Real-time Sentiment Analysis:** Adding a real-time analysis feature to the project could be valuable for investors. This could provide live updates on sentiment trends, enabling more immediate insights into how financial news impacts stock prices.
4. **Sentiment and Stock Movement Prediction:** The next step could involve using sentiment data to predict stock movements, potentially integrating machine learning models that combine historical stock data with sentiment to forecast future price trends.

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Appendices

1. DISSERTATION CODE: [https://drive.google.com/drive/folders/1V - HWUN8iiMEN3Teg5SV8dX0Ns-ynUYp?usp=drive_link](https://drive.google.com/drive/folders/1V-HWUN8iiMEN3Teg5SV8dX0Ns-ynUYp?usp=drive_link)
2. [Kaggle dataset](#)
3. Grammarly for correcting grammar: <https://www.grammarly.com/desktop>
4. Github Repositorie: <https://github.com/Faizahmad-S/stock-market-Sentiment-analysis.git>